

Direction and Residual Awareness Curriculum Learning Network for Rain Streaks Removal

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Abstract—Single-image rain streaks’ removal has attracted great attention in recent years. However, due to the highly visual similarity between the rain streaks and the line pattern image edges, the over-smoothing of image edges or residual rain streaks’ phenomenon may unexpectedly occur in the deraining results. To overcome this problem, we propose a direction and residual awareness network within the curriculum learning paradigm for the rain streaks’ removal. Specifically, we present a statistical analysis of the rain streaks on large-scale real rainy images and figure out that rain streaks in local patches possess principal directionality. This motivates us to design a direction-aware network for rain streaks’ modeling, in which the principal directionality property endows us with the discriminative representation ability of better differing rain streaks from image edges. On the other hand, for image modeling, we are motivated by the iterative regularization in classical image processing and unfold it into a novel residual-aware block (RAB) to explicitly model the relationship between the image and the residual. The RAB adaptively learns balance parameters to selectively emphasize informative image features and better suppress the rain streaks. Finally, we formulate the rain streaks’ removal problem into the curriculum learning paradigm which progressively learns the directionality of the rain streaks, rain streaks’ appearance, and the image layer in a coarse-to-fine, easy-to-hard guidance manner. Solid experiments on extensive simulated and real benchmarks demonstrate the visual and quantitative improvement of the proposed method over the state-of-the-art methods.

Index Terms—Convolutional neural network (CNN), curriculum learning, directionality, iterative regularization (IR), rain streaks’ removal, residual.

I. INTRODUCTION

THE rainy weather severely degrades the visual quality of the images, including the nearby rain streaks and distant veiling artifacts [1]. The rain would unexpectedly reduce the visibility of outdoor vision systems and bring negative effects to high-level vision tasks, such as scene classification [2], object detection [3], and semantic segmentation [4]. Numerous methods for different tasks such as the rain streaks’ removal [5] and jointly dehazing and deraining [6], [7], [8] have been proposed in recent years. In this work, we focus on

the single-image rain streaks’ removal via a simplified linear model

$$Y = X + B + N \quad (1)$$

where $Y \in \mathbb{R}^{R \times C}$ is an observed rainy image, R, C denotes the number of row and column, respectively, $X \in \mathbb{R}^{R \times C}$ represents the desired clean image, $B \in \mathbb{R}^{R \times C}$ means the rain streaks, and N is the random noise.

Recent works have made great progress in rain streaks’ removal and can be classified into three main categories by different strategies of incorporating knowledge. The core idea of the first one is to estimate the clean image or the rain streaks only from the rainy one [9], [10], [11], [12], [13], [14], [15], [16]. For example, Li et al. [13] and Ren et al. [15] designed a multistage progressive recurrent network to learn the rain streaks or the clean image in a supervised way, respectively. Such a straightforward manner individually models the image component and the rain streaks and may lead to less discriminative feature representation. The second one originates from the image decomposition viewpoint, in which both the image and the rain streaks are explicitly exploited. The pioneer works [17], [18], [19], [20], [21], [22] usually make use of the hand-crafted priors to model each component via the MAP-based optimization. These -representation-based sparse coding [17] or low-rank [17] methods are less effective for complex rain appearance. In recent years, the learning-based methods [23], [24], [25], [26], [27], [28], [29] with powerful representation have been naturally introduced with promising results. Li et al [25] jointly learned the image and rain via the decomposition and composition network.

To further improve the feature discrimination, the third category additionally introduces the side information, such as the density [11], location [23], [26], direction [5], [21], multi-scale [30], [31] of the rain streaks, or the depth [7], distribution [32], [33] of real rainy image. For example, Zhang et al. [11] proposed a density-aware convolutional neural network (CNN) for automatically sensing the rain density. From the image modeling perspective, Hu et al. [7] learned depth-aware features via a depth-guided attention mechanism for better differentiating the nearby rain streaks from the distant haze.

Although the existing methods have achieved great progress in the past decades, due to the highly visual similarity between the rain streaks and line pattern image edges, the results may still suffer from over-smoothing or residual rain streaks. Therefore, a crucial challenge for rain streaks removal is how to learn robust and discriminative representations so as to make the rain streaks and image layer distinguishable lie on two separable subspaces. To this end, we propose a novel

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direction and residual awareness network (DRAN) within the curriculum learning paradigm for single-image rain streaks' removal, which has several core advantages in rain streaks' modeling, image layer modeling, and network design.

A. Rain Streaks' Modeling

Most of the previous methods model the rain streaks via the CNN or hand-crafted prior. Nevertheless, it is still challenging to well differentiate the rain streaks from similar line pattern image structures. Previous filtering-based [34], [35] and optimization-based methods [5], [21], [36] have shown that directionality is beneficial to the rain streaks' removal, which motivates us to explore the directionality of the rain streaks in the learning framework.

In this work, we first conduct a statistical analysis of the rain streaks on large-scale real rainy images [37], and prove that natural rain streaks in local patches possess a principal direction (Section III-B1). Therefore, we design a direction-aware network (DAN) encoding the angle information to guide the rain streaks' estimation and illustrate how the auxiliary directionality knowledge endows DAN with the discriminative representation ability to adaptively distinguish rain streaks from image edge with different directions (Section III-B2). Specifically, the DAN consists of a principal direction regression subnetwork and a rain streaks' estimation subnetwork, in which the estimated angle of the rain streaks is imported into rain streaks' estimation subnetwork to differ the rain streaks from the image structure with other angles (Section III-B3).

B. Image Layer Modeling

The existing learning-based rain streaks' removal methods usually use the convolution or ResNet [2] to model the features of the image layer. In recent years, more advanced feature extraction layers have been proposed [38], [39]. In this work, we elaborately present a novel residual-aware block (RAB) as a simple and effective module for informative features' representation, which unfolds the iterative regularization (IR) [40] into a residual-aware network (RAN) (Section III-C2). The intuition behind IR is that the restored image in the $(k+1)$ th iteration is a weighted sum of the k th iteration and the residual between the two iterations, so as to retrieve the residual image structures back to the next iteration with better structure-preserving (Section III-C1).

Consequently, the RAB can not only explicitly model image feature but also encode the residual information with learned parameters as an attention mechanism to adaptively retrieve image structure feature and discard the rain streaks' feature. All the parameters and variables involved in the IR can be adaptively learned in an end-to-end manner. We analyze how RAB enhances feature representations with interpretability. Interestingly, we discover that RAB could offer a unified understanding of the ResNet [2] (Section III-C3). The architecture and loss for RAN are described in Section III-C4.

C. Curriculum Learning

The curriculum learning [41], [42], [43] is to learn the samples that are not randomly presented but organized in a meaningful order, in which a model is learned gradually from easy to complex samples. Usually, the predictions conform to a hierarchical coarse-to-fine structure in the output space. In this work, we formulate our whole network into a curriculum learning paradigm (Section III-A).

Naturally, we progressively learn the direction of the rain streaks, the rain streaks' layer, and the image layer, which correspond to the coarse scalar, sparse vector (each rain streak can be simplified as a combination of line pattern vectors), and dense matrix, respectively. The coarse-to-fine representation could significantly ease the training procedure. More importantly, these predictions are tightly coupled. The early prediction of the direction prior knowledge is beneficial to refine the rain streaks with principal direction. The latter prediction of the sparse rain streaks via the DAN could offer guidance about the location and appearance information which functions as the conditional attention map for better image estimation with the RAB.

The overall contributions are summarized and the innovations are highlighted as follows.

- 1) We first analyze the principal directionality property of the real rain patches and incorporate the angle side information into the learning network. Consequently, a direction-aware subnetwork is presented, which enables us adaptively inferring the angle of rain streaks and significantly enriches the discriminative features to differentiate the rain streaks from the image structures.
- 2) We unfold the classical IR into a novel RAN with enhanced feature representation ability for better image structure-preserving. Moreover, we provide a unified understanding between the ResNet and the RAB, which would enlighten the researchers for other low-level vision tasks.
- 3) A novel DRAN is proposed within the curriculum learning paradigm. The progressive guidance rain streaks' removal architecture learns the angle of the rain streak, rain streak layer, and the image layer in a coarse-to-fine and mutual guidance manner, which is of great benefit to improve the feature representation and alleviate the training difficulty.
- 4) We simulate a rain streaks' dataset with angle, rain streaks, and image layer labels. The proposed method has been evaluated on extensive simulated and real rain streaks' datasets and consistently outperforms the state-of-the-art methods both quantitatively and visually. Moreover, we validate the rain streaks' removal performance on different high-level tasks: image-level recognition, object-level detection, and pixel-level segmentation.

II. RELATED WORK

In Table I, we list the representative rainy image restoration methods and their main features. The interested reader could refer to survey work [64] and [65] for detailed description.

A. Image Deraining Tasks

The bad weather, e.g., haze [66], rain, or snow, severely degrades the visual quality of the images or videos and consequently high-level vision tasks. The bad weather [1], [67] can be further classified into the dynamic weather, where the constituent particles are of relatively large sizes (0.1–10 mm) such as the rain and snow, and the steady weather, where the constituent particles are of relatively small sizes (1–10 μ m) such as the haze. In this article, we focus on the rain, a common dynamic weather condition.

TABLE I
ILLUSTRATION OF THE EXISTING RAINY IMAGE RESTORATION METHODS

Paradigm	Method	Year	Input	Taks	Estimation	Label	Motivation/Contribution	Side information	Methodology
Physical Models + Filtering Methods	Garg [1]	2007	Video	Rain detection	Rain location	No	Comprehensive physical property and appearance model analysis of the rain	Size, Direction, Photometric, Shape, Temporal correlation	Linear photometric detection + Temporal filtering
	Barnum [34]	2010	Video	Rain detection	Rain location	No	Discover the invariance property of the rain streaks in frequency space	Shape, Direction, Breadth, Diameter, Depth	Shape and appearance model in frequency domain
	Bossu [35]	2011	Video	Rain detection	Rain location	No	Devise a novel hand-crafted rain feature: histogram of orientations of streaks	Photometric, Direction, Size, Temporal correlation	Photometric coarse detection + Histogram of orientations of streaks refine
	Kim [44]	2013	Single	Rain streaks removal	Rain location	No	Parameterize the rain streaks as elongated elliptical with two variables	Shape, Direction, Luminance	Shape and orientation detection + Nonlocal means filtering
	Chen [45]	2014	Video	Rain streaks removal	Rain location	No	Handle heavy rain in dynamic scenes, instead of the static background	Photometric, Chromatic	Photometric and chromatic detection + spatial-temporal local averaging
Image Decomposition based Optimization Methods	Kang [17]	2012	Single	Rain streaks removal	Clean image, Rain streaks	No	Firstly formulate the deraining task as an image decomposition problem	Sparsity	Image decomposition based dictionary learning for both image and rain streaks
	Chen [18]	2013	Single	Rain streaks removal	Clean image, Rain streaks	No	Reveal similar and repeated patterns of rain streaks and model with the low-rank	Sparsity, Self-similarity	Low-rank for rain streaks + Total variation for image
	Luo [19]	2015	Single	Rain streaks removal	Clean image, Rain streaks	No	Improve mutual exclusivity discriminative representation between the rain and image	Sparsity	Mutual exclusivity dictionary learning for both image and rain streaks
	Li [20]	2016	Single	Rain streaks removal	Clean image, Rain streaks	No	Introduce GMM for modeling multiple orientations and scales of rain streaks	Sparsity	Gaussian mixture models for both image and rain streaks
	Zhu [21]	2017	Single	Rain streaks removal	Clean image, Rain streaks	No	Take the directionality of rain streaks into consideration via gradient constraint	Sparsity, Direction	Dictionary learning and direction constraint for image + Weighted TV for rain streaks
	Chang [5]	2017	Single	Rain streaks removal	Clean image, Rain streaks, angle	No	Take the directionality of rain streaks into consideration	Sparsity, Direction, Self-similarity	Low-rank and Unidirectional TV for image + Transformed low-rank for rain streaks
	Jiang [36]	2017	Video	Rain streaks removal	Clean image, Rain streaks	No	Take the directionality of rain streaks into consideration for the video deraining	Gradient sparsity, Spatiotemporal correlation	Unidirectional TV and Low-rank for image + Unidirectional TV for rain streaks
	Ren [46]	2017	Video	Rain streaks removal	Background, Foreground, Rain streaks	No	Further decompose the clean image into static background and moving foreground	Sparsity, Spatiotemporal correlation	Low-rank for static background + MRF for moving targets
	Wei [47]	2017	Video	Rain streaks removal	Background, Foreground, Rain streaks	No	Encodes rain streaks in a stochastic way via patch-based mixture of Gaussians	Sparsity, Spatiotemporal correlation	Low-rank for background + spatiotemporal smoothness for moving object + MoG for rain
	Fu [9]	2017	Single	Rain streaks removal	Rain streaks	Yes	Introduce the deep residual CNN into the image deraining area	External dataset	Residual CNN
Learning based CNN Methods	Yang [23]	2017	Single	Rain streaks removal	Rain streaks, location, Clean image	Yes	Consider the rain accumulation effect and additionally model the location of the rain	Location, External dataset	Joint detection and restoration via multi-task learning
	Li [12]	2018	Single	Rain streaks removal	Rain streaks	Yes	Introduce non-local block into rain streaks removal for better feature expression	External dataset	Non-locally enhanced encoder-decoder network
	Qian [48]	2018	Single	Raindrops removal	Clean image	Yes	Inject visual attention into both the generative and discriminative network	Location, External dataset	Attention-recurrent network with adversarial learning
	Zhang [11]	2018	Single	Rain streaks removal	Clean image	Yes	Consider the different levels of the rain streaks via the density-aware network	Density, External dataset	Density-aware multiscale dense network
	Liu [49]	2018	Video	Rain streaks removal	High-frequency, Image, Rain streaks, Occlusion	Yes	Classify rain into streak and occlusion with adaptive removal or reconstruction strategy	Location, External dataset Temporal correlation	Joint recurrent rain removal and reconstruction network
	Chen [50]	2018	Video	Rain streaks removal	Clean image	Yes	Focus on real complex dynamic rainy scenes with opaque occlusions	Spatiotemporal correlation, Chromatic, External dataset	Superpixel segmentation-based alignment + Temporal filtering + CNN compensation
	Li [13]	2018	Single	Rain streaks removal	Rain streaks	Yes	Gradually removal of different rain streaks in multistage with SENet-bade RNN	External dataset	Multi-stage progressive recurrent network
	Liu [51]	2018	Single	Rain streaks removal	Clean image, Rain streaks	Yes	Combine the optimization-based method and CNN via plug-and-play framework	Physical model, External dataset	Total variation for image + CNN for rain streaks
	Wang [52]	2019	Single	Rain streaks removal	Clean image	Yes	Achieve better balance between removing rain streaks and reserving realistic details	External dataset	Alternatively training with CNN for details preserving and GAN for rain streaks removal
	Hu [7]	2019	Single	Rain and haze removal	Depth, Rain streaks	Yes	Take haze into consideration with depth side information for realistic rain removal	Depth, External dataset	Depth-attentional CNN
	Li [6]	2019	Single	Rain streaks removal	Depth, Rain streaks, Atmospheric transmission	Yes	Consider the veiling effect of heavy rains with physical-based GAN model	Atmospheric scattering model, External dataset	Physics-based CNN + Refine network + Adversarial network
	Ren [15]	2019	Single	Rain streaks removal	Clean image	Yes	Analyze the effectiveness of various modules with a new simpler baseline	External dataset	Multi-stage progressive recurrent resNet
	Wei [32]	2019	Single	Rain streaks removal	Rain streaks	Semi	Bridge the gap between simulated and real rainy image via semi-supervised learning	Gradient sparsity, External dataset	CNN for simulated rainy image + Total variation for real rainy image + KL divergence
	Wang [37]	2019	Single	Rain streaks removal	Clean image	Yes	Contribute the first real rain/clean image pairs for rain removal community	Real rain dataset	Spatial attentive CNN
	Yasarla [26]	2019	Single	Rain streaks removal	Rain streaks	Yes	Take the location and multi-scale property of rain streaks into consideration	Location, External dataset	Attentive multi-scale residual learning
	Zhu [26]	2019	Single	Rain streaks removal	Clean images	Semi	Bridge the gap between simulated and real rainy image via adversarial learning	Location, Unpaired dataset	Adversarial model with multiscale attention for both generator and discriminator
	Quan [53]	2019	Single	Raindrops removal	Clean image	Yes	Utilize the physical shape prior of the raindrops to enhance feature representation	Shape, External dataset	Shape and channel attention network
	Wang [16]	2019	Single	Rain streaks removal	Clean image	Yes	Enhance the low-quality embedding of classical UNet via entangled representation	External dataset	Enhanced encoded network
	Yang [54]	2019	Video	Rain streaks removal	Rain streaks, Accumulation, Occlusion, Atmospheric transmission	Yes	Formulate video derain into a two-stage framework: realistic single image derain + frame flow consistency derain	Atmospheric scattering model, Occlusion, Motion consistency	Physical-model based CNN for single image + Dual-level flow CNN transferring for video
	Yang [14]	2019	Single	Rain streaks removal	Clean image	Yes	Try to solve the scale inconsistency issue of the rain streaks via the wavelet	External dataset	Multi-scale based recurrent wavelet CNN
	Wang [28]	2020	Single	Rain streaks removal	Clean image, Rain streaks	Yes	Unfold the model-based decomposition method into interpretable deraining network	Physics model, External dataset	Model-driven deep convolutional dictionary network
	Du [55]	2020	Single	Rain streaks removal	Clean image, Rain density	Yes	Introduce the conditional variational auto-encoder into the derain community	Density, External dataset	Conditional variational auto-encoder with spatial density attention
	Liu [27]	2020	Single	Rain streaks removal	Clean image, Rain streaks	Yes	Combine the optimization-based method and CNN via plug-and-play framework	Physics model, External dataset	Model-driven deep CNN
	Fu [56]	2020	Single	Rain streaks removal	Clean image	Yes	Design the derain network with few parameters for the mobile devices	External dataset	Light weight pyramid networks
	Yang [57]	2020	Video	Rain streaks removal	Clean image	No	do not require any clean groundtruth images in the training process	Identical assumption of adjacent frames	Prediction network predicts current frame + Enhancement network compensates the details
	Deng [29]	2020	Single	Rain streaks removal	Clean image, Rain streaks	Yes	Compromise opposite goals: rain streaks removal and texture details preserving	External dataset	Two-stream parallel sub-networks
	Jiang [31]	2020	Single	Rain streaks removal	Rain streaks	Yes	Utilize the multi-scale property of the rain streaks	External dataset	Multi-scale progressive fusion network
	Yasarla [33]	2020	Single	Rain streaks removal	Clean image	Semi	Generalize the synethetical rain streaks to real-world rainy images	External dataset	Gaussian process-based semi-supervised learning
	Quan [58]	2021	Single	Rain drops and streaks	Clean image	Yes	Remove rain streaks and raindrops in a unified framework	External dataset	Complementary cascaded deraining network with neural architecture search
	Ye [59]	2021	Single	Rain streaks removal	Clean image, Rain streaks	Semi	Jointly learn real rain generation and removal within a unified framework	External dataset	Bidirectional disentangled translation network
	Huang [60]	2021	Single	Rain streaks removal	Clean image	Semi	Proposed a memory module to record various appearances of rain	External dataset	Memory-oriented semisupervised method with self-training mechanism
	Yan [61]	2021	Video	Rain and haze removal	Clean image	Yes	Employ deformable convolution layers to avoid explicit optical flow	External dataset	Feature align for rain streak removal and transmission-depth consistency for haze
	Liu [62]	2021	Single	Rain streaks removal	Clean image	Semi	Utilize the real rain image with a rain direction regularizer	Rain direction, External dataset	Unpaired semi-supervised learning with knowledge distillation
	Wei [63]	2021	Single	Rain streaks removal	Clean image	No	Remove the rain streaks in an unsupervised way	Rain direction, External dataset	Joint rain removal and generation within the CycleGAN framework

The physical shape of the raindrop is almost spherical for the small size drop and oblate spheroid for larger size [1]. Due to the exposure time of the imaging sensor and moving property of the raindrops, the appearance of the rain in the

image space is usually the line pattern rain streaks, which can be interpreted as a motion blurring procedure of the raindrops. Thus, the rain can be further classified into the raindrop [48], [53], [68] and rain streaks [5], [17], [20], [21].

From the image/video perspective, the image deraining tasks can be categorized as the single-image deraining and video deraining [34], [35], [36], [45], [47], [49], [50], [68], [69], [70], [71], [72]. The video deraining methods could further use the temporal redundancies among the sequences. In recent years, the researchers begin to pay attention to the realistic rain model where the rain and the haze always coexist [6], [7], [73]. In this work, we mainly focus on the single-image rain streaks' removal task. Our goal is to explore the key issue in rain streaks' removal: how to learn discriminative feature for the image and rain streaks, so as to better differentiate the rain streaks from the visual similar line pattern image edges.

B. Rain Streaks' Removal Methods

The goal of image rain streaks removal is to separate the rain streaks from the clean image. From the tool viewpoint, the existing methods can be mainly classified into the following categories: the physical models [1], [67], the filtering-based image processing methods [34], [44], [45], the model-based optimization [5], [17], [18], [19], [20], [21], [22], and the learning-based CNN [6], [7], [9], [10], [11], [12], [13], [14], [15], [16], [23], [24], [26], [32], [37], [74], [75]. The model-based optimization methods usually use the decomposition framework to model the image and rain streaks simultaneously with hand-crafted priors, such as sparse coding [17], [19], [21], GMM [20], or low-rank [5], [18]. For example, Luo et al. [19] proposed a discriminative sparse coding method by regularizing the two learned dictionaries with mutual exclusivity property, thus making the estimated rain and image layer distinguishable. Li et al. [20] proposed a Gaussian mixture models (GMMs) for both the rain layer and the image layer with impressive performance.

Although the optimization-based methods have achieved excellent rain streak removal results, these hand-crafted priors are less robust to handle the rain streaks with diverse distributions, due to the varied angle, location, intensity, density, length, width, and so on. The CNN-based methods can implicitly learn the rain streak priors via the powerful representation ability along with the rain streaks' datasets. Fu et al. [9] introduced the end-to-end residual CNN to solve the rain streaks' removal problem. Li et al. [6] presented a physics-based backbone followed by a depth-guided GAN refinement for heavy rain image restoration. Ren et al. [15] proposed a simple yet effective progressive recurrent network with recursive blocks for image deraining. In this work, our progressive learning framework is in line with [15]. However, we further formulate the progressive network in a hierarchical coarse-to-fine structure via curriculum learning.

C. Side Information

Side knowledge is exploited to enhance the discriminative features for better differing rain streaks from the image layer. The intuitive knowledge about the rain streaks have been widely considered: location [23], [48], [76], density [11], shape [44], and direction [35]. There are also useful prior knowledge for image layer, such as depth [7] and distribution of real rainy image [32]. For example, Yang et al. [23] learned the location and appearance of rain streaks to guide the rain image restoration. Similar location and shape priors can be naturally applied in raindrops' removal [48], [53]. Zhang et al. [11] enabled the network to sense the rain density and removed the rain streaks guided by the density. Hu et al. [7] designed

a depth-attention network to handle both the rain and the haze.

The directionality of the rain streaks has been already used in previous deraining methods. Barnum et al. [34] analyzed the properties of rain streaks in frequency space and enforced a directional component filtering in the Fourier-transformed domain. Bossu et al. [35] proposed a histogram of orientations of the rain streaks method along with the photometric property to detect the rain streaks. Latter, Chang et al. [5], Zhu et al. [21], and Jiang et al. [36] further used the directionality of rain streaks in the optimization framework. In this work, we first analyze the directionality of the real rain streaks and propose the direction-aware network to validate the effectiveness of the directionality in neural network.

III. DIRECTION AND RESIDUAL AWARENESS NETWORK

A. Curriculum Learning Framework of DRAN

In this article, we develop a DRAN within the curriculum learning framework for single-image rain streaks' removal. The overview architecture of DRAN is shown in Fig. 1. For rain streaks' modeling, we design a direction-aware network to encode the angle information for boosting rain streaks' estimation. More specifically, we learn the angle of the rain streaks in a supervised manner, and then feed the embedded angle along with the input into the rain streaks' estimation module to enhance the feature representation. For image layer modeling, we devise the RAN for image estimation by stacking RABs, specialized to adaptively model the relationship between the image and the residual. Finally, the estimated rain streaks concatenated with the original input are fed into the image estimation network, so that the network could pay more attention to the rain streaks' region.

On one hand, the learning samples are structurally organized in a meaningful order from the principal direction of the rain streaks' angle, rain streaks' visual appearance, and clean image. The deraining results of DRAN gradually approach to the clean image from the scalar-level angle, to the sparse vector-level rain streaks, and finally to the dense matrix-level image layer. Such a coarse-to-fine and simple-to-complex guidance network would significantly reduce the training difficulty and increase the feature representation. One the other hand, the hierarchical information estimated via the subnetworks in DRAN are tightly coupled in a progressive guidance manner. The estimated angle information benefits to differentiate the estimated rain streaks from the image edges with other directions. The estimated rain streaks would provide abundant location/appearance guidance for better separating the rain streaks from the image content.

B. Direction-Aware Network for Rain Streaks

In this section, we first analyze the rationality of the directionality of the real rain from both the imaging mechanism and the statistical perspective. Then, we further illustrate the benefit of the directionality of the rain streaks. Finally, we describe the details of the architecture and loss for the DAN about how we use the directionality of the rain streak.

1) *Rationality of Principal Direction*: First, we briefly introduce the physical properties of rain. The physical shape is spherical for small radius raindrop and oblate spheroid for large radius raindrop [1]. With the coaction of gravity and wind, the raindrop reaches a terminal velocity. However, due

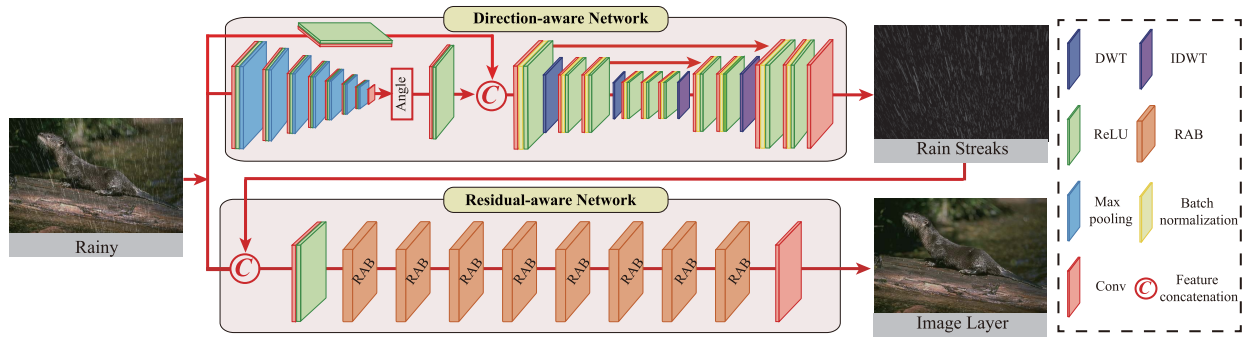


Fig. 1. Overview of the proposed DRAN, which is composed of two networks: direction-aware network for rain streaks and residual-aware network for image modeling. The DAN integrates the direction knowledge to differentiate the rain streaks from the image edges better; the RAN consists of several RABs to adaptively model the relationship between the image and the residual. Moreover, the progressive guidance from the angle to rain streaks, and then from rain streak to the image would further enhance the feature representation.

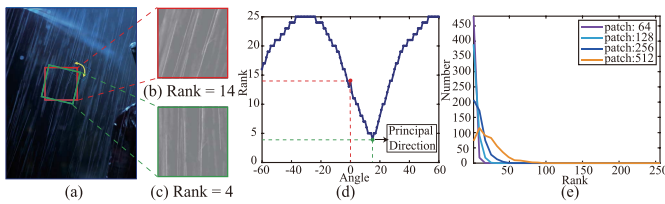


Fig. 2. Statistical analysis of the directionality of real rain streaks. (a) Rainy, (b) original patch, and (c) rotated patch. (d) Rank changing curve with different rotation angles. (e) Histogram of the minimal rank of real rain streaks with different sizes.

to the long exposures of the imaging system and fast motion of the raindrops, the visual appearance of the rain in the image space is presented as severely motion-blurred rain streaks, as shown in Fig. 2(a) a real rainy image in [37]. It is intuitively observed that the rain streaks exhibit obvious directionality.

We further quantitatively demonstrate that the regular line pattern rain streaks in the local patch share a principal direction. To illustrate this, we use the principal component analysis (PCA) to quantitatively analyze the rank of the rain streaks. We gradually rotate the original rain streaks [red patch with rank = 14, Fig. 2(b)] to search its lowest rank. We discover that the rotated green rain streaks' patch shows clearly vertical direction with rank = 4 in Fig. 2(c). Furthermore, in Fig. 2(d), we plot the rank changing curve with different rotation angles of Fig. 2(b). We can observe that there exists minimal rank with a certain angle. *The lower rank the rain streaks' patch is, the clearer the principal direction is.*

Finally, we conduct a statistical optimal minimal rank analysis experiment on 4000 real rain streaks [37] with different patch sizes. In Fig. 2(e), the horizontal index denotes the minimal rank of rain streaks, and the vertical index means the corresponding numbers. The heavy-tailed sparse distributions indicate that most of the rain streaks patch have distinct low-rank property when the patch size is smaller than 256×256 . The analysis reveals an intuitive yet important fact that the principal direction characteristic holds for most of the real rain streaks.

Considering there is no rain dataset with the angle label of the rain streaks, we follow the conventional rain streaks' simulation method [9] and construct a training dataset with 15,000 outdoor scene images from place2 [77] and a test dataset with 200 outdoor images, named as rain200Place. In particular, each synthetic rainy image is assigned with a

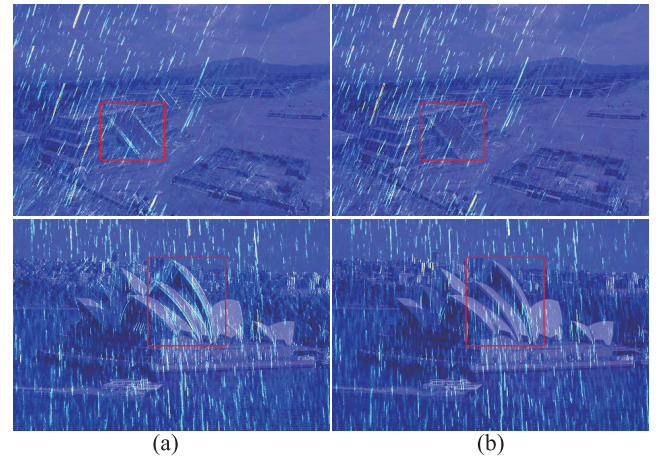


Fig. 3. Effectiveness of the direction knowledge for rain streaks' estimation. We visualize two example features via the heatmap (a) without and (b) with the direction guidance.

direction label, a rain streak label, and an image label. The direction label of the rain streaks is very easy to know in advance. *It is worth noting that the angle of the rain streaks in one image may not be exactly the same. Thus, we set a small variance around one specific angle for each rainy image so as to improve the robustness for real rain streaks.*

2) *Benefit of Directionality:* Most existing CNN-based methods [10], [13], [15], [32] directly learn the mapping from the rainy image to the rain streaks, which mainly exploit the visual features of the rain streaks. However, this may be difficult to differentiate the rain streaks from the line pattern image edges, due to high visual similarity. We alleviate this problem with a novel DAN [Fig. 1] by incorporating the angle auxiliary knowledge, which could enable the network to perceive the specific orientation features and pay more attention to the rain streaks around the principal direction.

To illustrate how directional feature may help improve the discrimination, instead of directly displaying the rain layer estimation results, we visualize the rain-related features and demonstrate how the DAN influences the learned rain-related features. We extract the rain features' heatmap of the last layer in DAN (one layer before the supervised rain layer), since they are mostly closed to the estimated rain streaks and physically meaningful. In Fig. 3, we visualize the rain features of two examples in the Rain12 dataset [20]: top

row for the *Relic* and bottom row for the *Opera House* and cover the rain feature heatmap onto the original image for better comparison between the rain features and the image edges. The left and right columns denote the heatmap with and without angle guidance, respectively. When the explicit angle annotation is provided, the learned features near the output are more discriminative, which mostly response in the rain streak regions, as shown in Fig. 3(b). On the contrary, when the angle annotation is not provided, the feature response contains considerable line pattern image edges in Fig. 3(a). These results intuitively demonstrate that the directionality in DAN could effectively improve the feature discrimination for better differing the rain streaks from the image edges.

3) *Architecture and Loss for DAN*: The DAN contains two mainly subnetworks: rain angle estimation subnetwork and rain streak layer estimation subnetwork, as shown in Fig. 1. We take the rain angle estimation as a regression task between 0° and 180° . Concretely, we design a direction regression network (seven CRP layers and one Conv layer, where CRP denotes the Conv + Relu + Pooling) to obtain the principal direction of the rain streaks, namely, the scalar angle information. And we stretch the scalar angle to the orientation feature with the same size as the original image. Then along with the rainy image, the stretched angle functioning as a conditional map is fed to the 16-layer U-Net-based rain streaks' estimation subnetwork. It is worth noting that we replace the conventional downsampling Conv and upsampling Deconv layer with the discrete wavelet transform (DWT) and inverse discrete wavelet transform (IDWT) layer. Since the DWT and IDWT are invertible, we can preserve useful information and enlarge receptive field for the context. In this work, we choose the Haar wavelet. Thus, the loss for the DAN contains both the rain angle and the rain layer

$$L_{\text{rain}} = \frac{\alpha}{2} \|\mathcal{F}_R([Y]) - \hat{\theta}\|^2 + \frac{\beta}{2} \|\mathcal{F}_A([Y, \theta]) - \hat{B}\|^2 \quad (2)$$

where θ , B , and Y denote the angle, the rain streak layer, and the rainy image, respectively, $\mathcal{F}_R$ and $\mathcal{F}_A$ are the subnetwork mapping functions for the angle and the rain streak, respectively, and α and β are the balance parameters.

C. Residual-Aware Network for Image Layer

In image decomposition-based rain streaks' removal, rain streaks' modeling and image layer modeling are the two sides of the same coin. We have designed the direction-aware network to enhance the discriminative features for rain streak modeling. Here, we further enhance the informative features for image layer modeling.

1) *Iterative Regularization*: The goal of image estimation is to increase its sensitivity to informative features so that we can well preserve the useful image details. To achieve this goal, we borrow the idea of IR [40] from classical image processing

$$B^{(k+1)} = B^{(k+1/2)} + \gamma (B^{(k)} - B^{(k+1/2)}) \quad (3)$$

where B^k is the intermediate result of the k th iteration, γ is a manually setting hyper-parameter, and $B^{(k+1/2)}$ is usually obtained by the threshold shrinkage method [78]

$$B^{(k+1/2)} = \text{shrink}(DB^{(k)} + J^{(k)}) \quad (4)$$

where shrink is the soft shrinkage operator for different priors [5], [19], D is the sparse transformation operator, and J is the adaptive compensation.

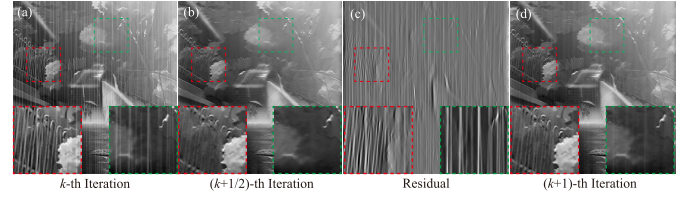


Fig. 4. Visualization of the iteration procedure of IR. (a) Current iteration result $B^{(k)}$, (b) intermediated result $B^{(k+1/2)}$, (c) residual error between (a) and (b), and (d) next iteration result $B^{(k+1)}$. The IR could retrieve the image structures, such as the fence marked by the red rectangle, while at the cost of the rain streaks marked by the green rectangle.

The basic idea of IR is twofold. On one hand, the restored intermediated image $B^{(k+1/2)}$ [the first term on the right of (3)] should keep the global similarity to last iteration result $B^{(k)}$; on the other hand, the unexpectedly lost image details [second term at right of (3)] in each iteration, due to the shrinkage operation, should be well-retrieved. In Fig. 4, we visually illustrate how the IR works. Given the input image [Fig. 4(a)], the shrinkage operator removes the rain streaks at the cost of unexpectedly image details lost [Fig. 4(b)]. We can observe that the residual $B^{(k+1/2)} - B^{(k)}$ between the shrinkage and the last iteration image contains both significant rain streaks (marked by green dashed rectangle) and image details (marked by red dashed rectangle), as shown in Fig. 4(c). Compared with Fig. 4(b), the IR could compromisingly retrieve important image structures, namely, the residual back to the next iteration [Fig. 4(d)].

2) *Residual-Aware Block*: However, the IR has one key issue. The hyperparameter γ functions as the balance between the restored image and the residual details. Such an important adaptive adjustment mechanism is just controlled by a manual scalar constant, which makes it difficult to differentiate the useful image detail from rain streak features as shown in Fig. 4(d). To mitigate this, we implement shrinkage operator (4) with a convolutional layer

$$B^{(k+1/2)} = \mathcal{F}(B^{(k)}) = S(W^{(k+1)} \otimes B^{(k)} + P^{(k+1)}) \quad (5)$$

where W , B , P , and S are the convolutional weights, intermediate layer features, bias, and activation function, respectively, and $\mathcal{F}$ is the convolutional function. Equations (4) and (5) are essentially similar sharing the same formulation: a linear transformation with a bias to extract features, followed by a nonlinear truncated function to activate the most useful features. Thus, we can reformulate (3) by replacing the shrinkage operator in (4) with CNN in (5)

$$B^{(k+1)} = \mathcal{F}(B^{(k)}) + \delta(B^{(k)} - \mathcal{F}(B^{(k)})). \quad (6)$$

It is worth noting that we replace the manual-defined scalar γ with the learnable vector δ , realized by spatially 1×1 convolution with 64 channels in the second conv layer of each RAB, as shown in Fig. 5(a). The δ representing the convolution operator can be regarded as the pointwise production with $B^{(k)} - \mathcal{F}(B^{(k)})$, with additional summation. The learnable vector δ has two distinct advantages over the fixed hyperparameter scalar γ : spatial and channel dimension. As for the conventional IR, the spatial value is fixed over the iteration, and the channel is only one for natural 2-D image. On one hand, from the spatial dimension, the learned balance parameter δ can be viewed as an attention mechanism to adaptively reallocate the informative residual features back to the input feature; on the other hand, from the channel dimension,

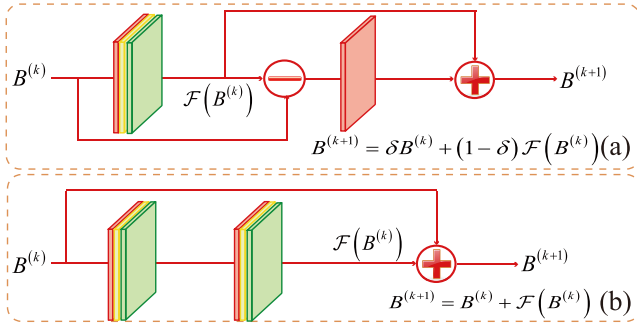


Fig. 5. Illustration of RAB and comparison with Resblock. The architecture and formulation of (a) RAB and (b) Resblock.

the abundant multiple channels in CNN significantly enrich the feature representation. We name (6) as the RAB, and design the network accordingly as shown in Fig. 5(a).

3) *Unified Understanding*: To further analyze the RAB, we reformulate (6) as

$$\mathbf{B}^{(k+1)} = \delta \mathbf{B}^{(k)} + (1 - \delta) \mathcal{F}(\mathbf{B}^{(k)}). \quad (7)$$

When δ is equal to 0.5,¹ the first term identity $\mathbf{B}^{(k)}$ and second term in (7) are treated equally. In such interesting case, the RAB and ResNet [2] share the intrinsic similarity with only a 0.5 scaling factor difference as shown in Fig. 5

$$\mathbf{B}^{(k+1)} = \mathbf{B}^{(k)} + \mathcal{F}(\mathbf{B}^{(k)}). \quad (8)$$

And when δ is equal to 0, the RAB degenerates to the classical convolution. The RAB (7) and ResNet (8) share the intrinsic similarity: the combination of the identity mapping and the residual feature. However, compared with the ResNet equally using the two terms, the RAB further learns an attention gating vector δ to adaptively balance the identity and residual feature. For rain streaks' removal task, δ could enhance the feature representation power for better differentiating the residual rain streaks from the image structures. To validate this, we compare the training error and PSNR between RAB and ResNet during the training procedure in Fig. 6. The parameters of ResNet and RAB are approximately equal. We can observe that the training errors of RAB are lower than those of ResNet, while the PSNR values are consistently higher than those of ResNet.

4) *Architecture and Loss for RAN*: The RAN is a simple plain network in Fig. 1. We stack eight RABs to form the residual-aware image estimation network. The input of RAN is the rainy image along with the estimated rain layer serving as a conditional map. The first convolutional layer in RAB consists of 64 filters with the size of $3 \times 3 \times 64$, and the second convolutional layer in RAB consists of 64 filters with the size of $1 \times 1 \times 64$. The loss for the RAN is the MSE loss

$$\mathcal{L}_{\text{image}} = \frac{\mu}{2} \|\mathcal{F}_I([Y, \mathbf{B}]) - \hat{\mathbf{X}}\|^2 \quad (9)$$

where $\mathbf{X}$ denotes the clean image, μ is the hyperparameter, and $\mathcal{F}_I$ is the subnetwork mapping functions for clean image. The overall loss of our method is $\mathcal{L} = \mathcal{L}_{\text{rain}} + \mathcal{L}_{\text{image}}$. We implement our framework on the TensorFlow platform and initialize the weights by [79] with Adam optimizer and set mini-batch size as 8. All the experiments are carried out on the PC with Intel

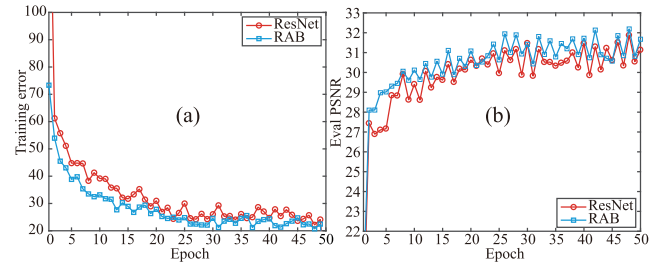


Fig. 6. Comparison between RAB and ResNet. (a) Training loss and (b) PSNR.

i7-7700 CPU and GTX 1080Ti, which takes about 3 days for training.

D. Training Details

We train the three subnetworks individually and then fine-tune the whole network on 256×256 local patches. For direction regression subnetwork, it takes 40 epochs with initialized learning rate $5e-4$, decayed to $1e-4$ (divided by 5) after ten epochs, and then decayed to $2e-5$ after 20 epochs. For rain streaks' estimation subnetwork, it takes 40 epochs with initialized learning rate $1e-3$, decayed to $2e-4$ after ten epochs, and then decayed to $4e-5$ after 20 epochs. For residual-aware image estimation subnetwork, the training setting is the same as that of the rain streaks. Finally, we fine-tune the whole network for 60 epochs. The learning rate is initialized with $5e-5$, decayed to $5e-6$ after 20 epochs, and then $1e-6$ after 40 epochs. The loss for rain angle, rain streaks, and image layer are supervised via $\mathcal{L}_2$. In the individual training stage, we equally set $\alpha = 1$ for rain angle, $\beta = 1$ for rain layer, and $\mu = 1$ for image layer. In the fine-tune stage of the overall network, we empirically set the hyperparameter $\alpha = 0$, $\beta = 0.1$, and $\mu = 1$. Note that $\alpha = 0$ in the overall fine-tune stage means that we will keep it fixed while we have learned the weights of the direction regression subnetwork in previous individual training stage. The pretrained models and training code will be available on our homepage.² Note that we have followed the recurrent strategy in JORDER-E [81] by cascading the proposed network three times with more powerful representation ability.

Compared with the previous supervised methods (image, rain streaks, or both of them), the proposed method additionally needs the direction information of the rain streaks. It is worth noting that the direction information is easy to obtain when we simulate the rain streaks via the physical degradation model of the motion deblurring procedure, namely, the angle of the blur kernel. Once we have learned the weights of the direction regression subnetwork on the proposed synthetic dataset, we will fix the weights of the direction regression subnetwork forever, since other rain datasets do not have the direction labels. When we test on other rain datasets, we only retrain and fine-tune our model with the rain streaks' estimation subnetwork and the residual-aware subnetwork. This is exactly the same as previous deraining methods.

IV. EXPERIMENTAL RESULTS

A. Datasets and Evaluation

We evaluate our DRAN on seven public synthetic datasets: 1) rain200Place where the direction of rain streaks slightly

¹We use the scalar for simplicity, which means each element in δ is all 0.5.

²<https://owuchangyuo.github.io/>

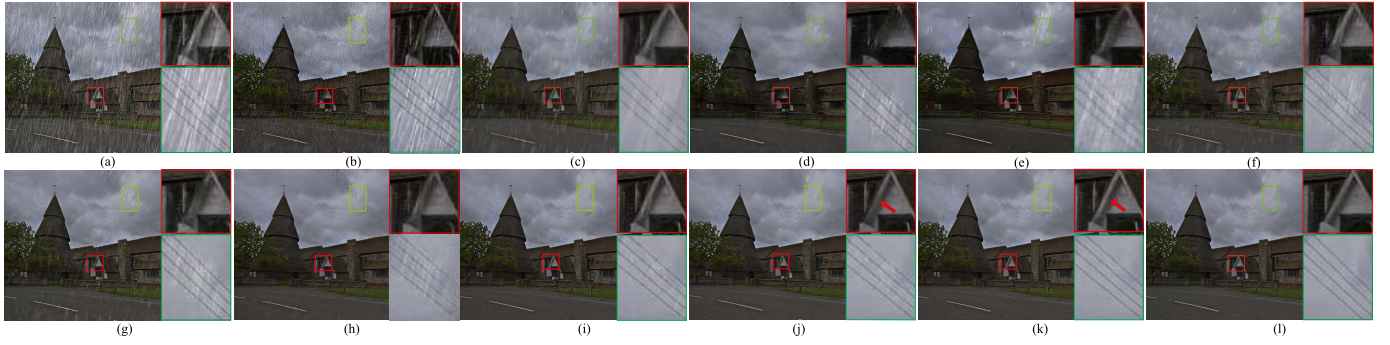


Fig. 7. Visual results of rain streaks' removal on the rain200Place dataset. (a) Rainy image. Deraining results by (b) DSC, (c) GMM, (d) DDN, (e) DID-MDN, (f) JORDER, (g) SPANet, (h) DRDNet, (i) PreNet, (j) JORDER-E, (k) DRAN, and (l) GT.

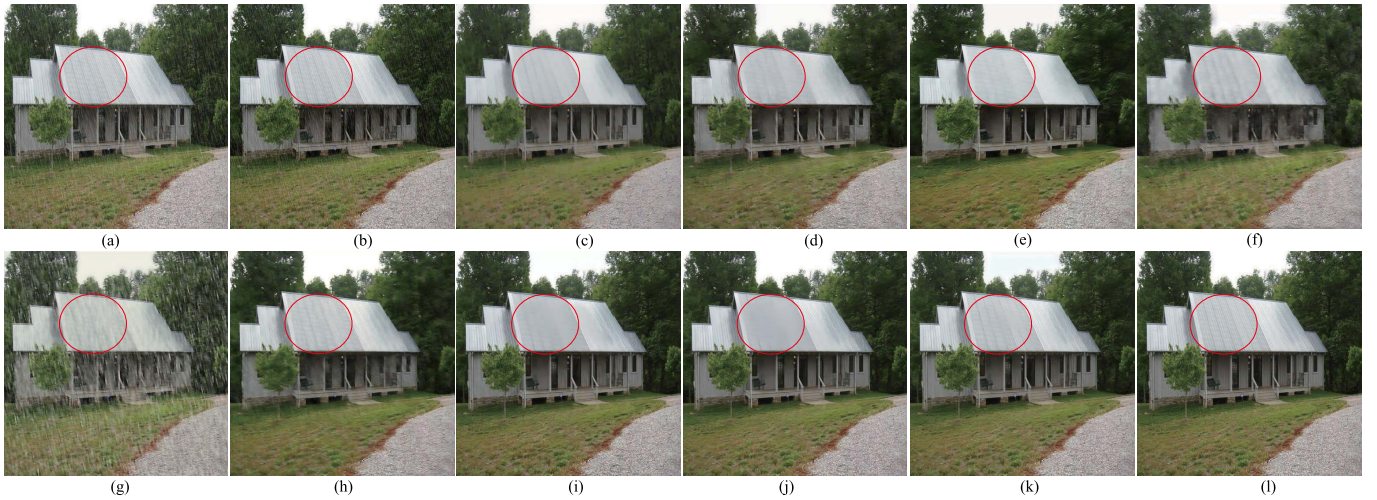


Fig. 8. Visual results of rain streaks' removal on the Rain12000 dataset. (a) Rainy image. Deraining results by (b) DSC, (c) GMM, (d) DDN, (e) DID-MDN, (f) JORDER, (g) SPANet, (h) DRDNet, (i) PreNet, (j) JORDER-E, (k) DRAN, and (l) GT.

varies in single image; 2) rain12 [20]; 3) rain1400D [9] includes 14 different angles of the rain streaks; 4) rain100L [23] and rain100H [23] with various rain layers; 5) Rain12000 [11] with different rain density; and 6) Rain800 [80]. It is worth noting that the direction, intensity, length, width, density, and distribution of rain streaks vary a lot. Moreover, we evaluate the proposed method on two real rain datasets: 1) RealHQRain [37]: a large-scale high-quality real rain dataset, which covers a wide range of natural rain scenes with rainy/clean image pair, and 2) Real100Rain: 100 real-world rainy images collected from the Internet.

For the evaluation index, we consider the PSNR and SSIM [82] for quantitative evaluation. The indexes of all the methods are calculated in RGB channels, not in the luminance channel. The proposed method and the competing methods follow exactly same setting. In fact, we have verified that the results of RGB space or the luminance space are very close to each other, within less 0.2 dB. We think it is more appropriate and reasonable to evaluate in the RGB space, since the input and output are originally in RGB space. Furthermore, we evaluate the deraining results on high-level vision tasks [65]: 1) image-level scene understanding (Google Vision API); 2) object-level detection [3]; and 3) pixel-level semantic segmentation [83].

B. Experimental Setting

We compare the proposed DRAN with six state-of-the-art methods: two optimization-based methods, i.e., DSC [19] and GMM [20]; seven learning-based methods, i.e., DID-MDN [11], DDN [9], JORDER [23], SPANet [37], DRD-Net [29], PreNet [15], and JORDER-E [81]. The DSC and GMM are unsupervised optimization-based methods, which can be used directly for any datasets. The trained models of DID-MDN, DDN, JORDER, and SPANet are downloaded from the homepage of the authors. For datasets with paired clean and rainy images, we will **retrain** the competing methods on the corresponding training datasets for fair comparison. Note that we learn the new model with random initialization, not fine-tune with the pretrained baselines. As for the real dataset Real100Rain, we would directly use the released baseline models of each method, since no ground truth (GT) is provided.

C. Evaluation on Synthetic Datasets

1) *Qualitative Evaluation*: Figs. 7 and 8 show simulated rain streaks' removal results on rain200Place and rain12000. Note that the rain streaks from different datasets have different visual appearances, such as direction slightly varied rain in rain200Place, high density rain in rain800, and high intensity

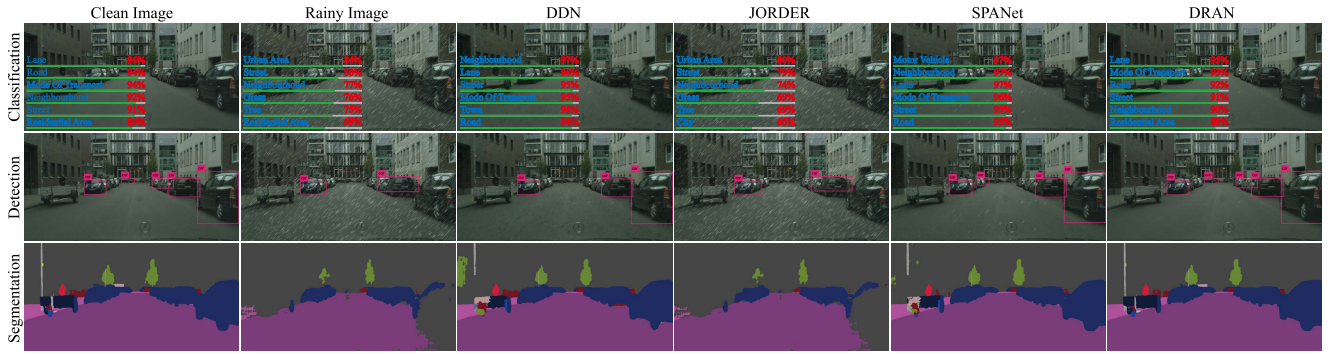


Fig. 9. Visual evaluation on high-level tasks: image-level recognition, object-level detection, and pixel-level segmentation from the first to third rows.

TABLE II
QUANTITATIVE COMPARISON ON TYPICAL SYNTHETIC DATASETS AND ONE REAL DATASET. BOLD MEANS THE BEST PERFORMANCE

Dataset	rain200Place		rain12 [20]		rain1400D [9]		rain100L [23]		Rain12000 [11]		Rain800 [81]		RealHDrain [37]		rain100H [23]	
Index	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Rainy	22.88	0.6801	28.82	0.8351	23.67	0.7564	25.52	0.8244	22.14	0.6863	21.13	0.6415	32.64	0.9315	12.13	0.3448
DSC [19]	24.95	0.6983	28.34	0.8397	26.14	0.7829	25.73	0.8271	24.88	0.7136	21.01	0.6444	32.33	0.9335	13.79	0.3575
GMM [20]	27.63	0.8429	30.70	0.8864	26.00	0.8013	27.01	0.8288	23.80	0.7487	20.84	0.7257	32.99	0.9475	12.55	0.3654
DDN [9]	30.59	0.9094	32.10	0.9302	28.37	0.8768	28.29	0.9034	29.14	0.8482	23.65	0.7863	28.96	0.9727	24.53	0.7498
JORDER [23]	26.61	0.8045	34.50	0.9434	26.23	0.8287	35.22	0.9687	25.52	0.8001	21.21	0.7358	35.72	0.9776	24.37	0.7855
DID-MDN [11]	25.84	0.8461	23.19	0.8338	27.76	0.8617	24.52	0.8181	25.84	0.8461	23.10	0.7726	28.96	0.9457	23.97	0.7748
SPANet [37]	29.47	0.9084	33.16	0.9416	27.81	0.8655	29.31	0.9205	24.90	0.7903	22.85	0.8114	38.06	0.9867	25.20	0.8378
PreNet [15]	33.56	0.9547	35.38	0.9547	30.47	0.9135	36.07	0.9749	31.54	0.9059	25.12	0.8439	35.17	0.9488	28.57	0.8866
DRD-Net [29]	30.80	0.9077	33.91	0.9310	29.50	0.8780	32.26	0.9351	29.29	0.8315	23.57	0.7794	34.24	0.9428	24.69	0.7887
JORDER-E [81]	33.90	0.9551	35.89	0.9595	30.70	0.9127	37.52	0.9804	31.99	0.9044	26.55	0.8556	40.67	0.9804	29.58	0.8883
DRAN	35.43	0.9663	36.02	0.9576	31.38	0.9189	37.83	0.9813	32.41	0.9107	27.31	0.8557	41.41	0.9805	30.65	0.9007

rain with various directions in rain100H. We have the following observations. First, the performance of our method is consistently good for different kinds of rain streaks, which verifies the robustness of the proposed method. Second, the proposed method consistently achieves better visual appearance results with respect to both the image details' preservation and rain streaks' removal. For example, the DRAN could well preserve the image line pattern edges and texture (such as the roof in Fig. 7). Moreover, for the accumulated wide rain streaks with high intensity, the DRAN leaves less residual rain streaks in the image background. Third, in Fig. 7, the directions of the rain streaks are not strictly the same. The proposed method has obtained satisfactory results which demonstrate the robustness to the rain streaks with slightly varied direction of different locations. These results strongly support the effectiveness of the DAN and RAB for better distinguishing the rain streaks from the image edges.

2) *Quantitative Evaluation*: Table II reports the PSNR and SSIM values of the competing methods. The simulated rain streaks of four different datasets vary a lot in terms of the direction, intensity, length, width, and density. We can observe that the proposed DRAN consistently outperforms the competing methods for all the datasets. The results further demonstrate the effectiveness of the proposed DRAN.

It is worth pointing out that the direction supervision is only a scalar, which is much weaker than that of the rain streaks (sparse vector) and the clean image (dense matrix). The direction prior is actually auxiliary side information to refine rain streaks' estimation. The final deraining results still heavily depend on rain streaks and image layer estimation. Nevertheless, we can note that the direction information benefits our model with 0.4 dB improvement. The direction supervision

is specially designed for differing the rain streaks from the similar line pattern image structures with other directions. The edges benefiting from the direction supervision as shown in Fig. 3 are the minority in an image. Thus, this improvement is very reasonable and impressive.

3) *High-level Task Evaluation*: Most of the existing methods evaluate the deraining performance via low-level visual assessments, such as the PSNR and SSIM. Recently, the high-level task evaluation has attracted more attention [65]. In this work, we evaluate the rain streaks' removal performance on three kinds of high-level vision tasks, including image-level scene understanding, object-level detection on VOC2007 [84] using YOLOv2 [3], and pixel-level semantic segmentation on Cityscape using PSPNet [83]. The qualitative evaluation results are shown in Fig. 9, and Table III reports the quantitative evaluation results. We have the following observations. First, compared with object detection, the semantic segmentation is much more influenced by the rain streaks. That is to say, the finer the vision task, the more serve the influence by rain streaks. Second, compared with other methods, the proposed DRAN obtains better visual results and quantitative index. Moreover, the results of DRAN are much closer to those of the GTs. Third, although directly training on the rainy image (Rainy+) has achieved very good results, the proposed DRAN still performs better than Rainy+.

D. Evaluation on Large-Scale Real Datasets

We test the proposed method on Real100rain and RealHQRain [37]. The visual deraining results of Real100rain and RealHQRain are shown in Figs. 10 and 11. Due to space limitation, we only show parts of the competing methods. In Fig. 10, DRAN outperforms the competing methods in terms of rain streaks' removal and detail-preserving. For

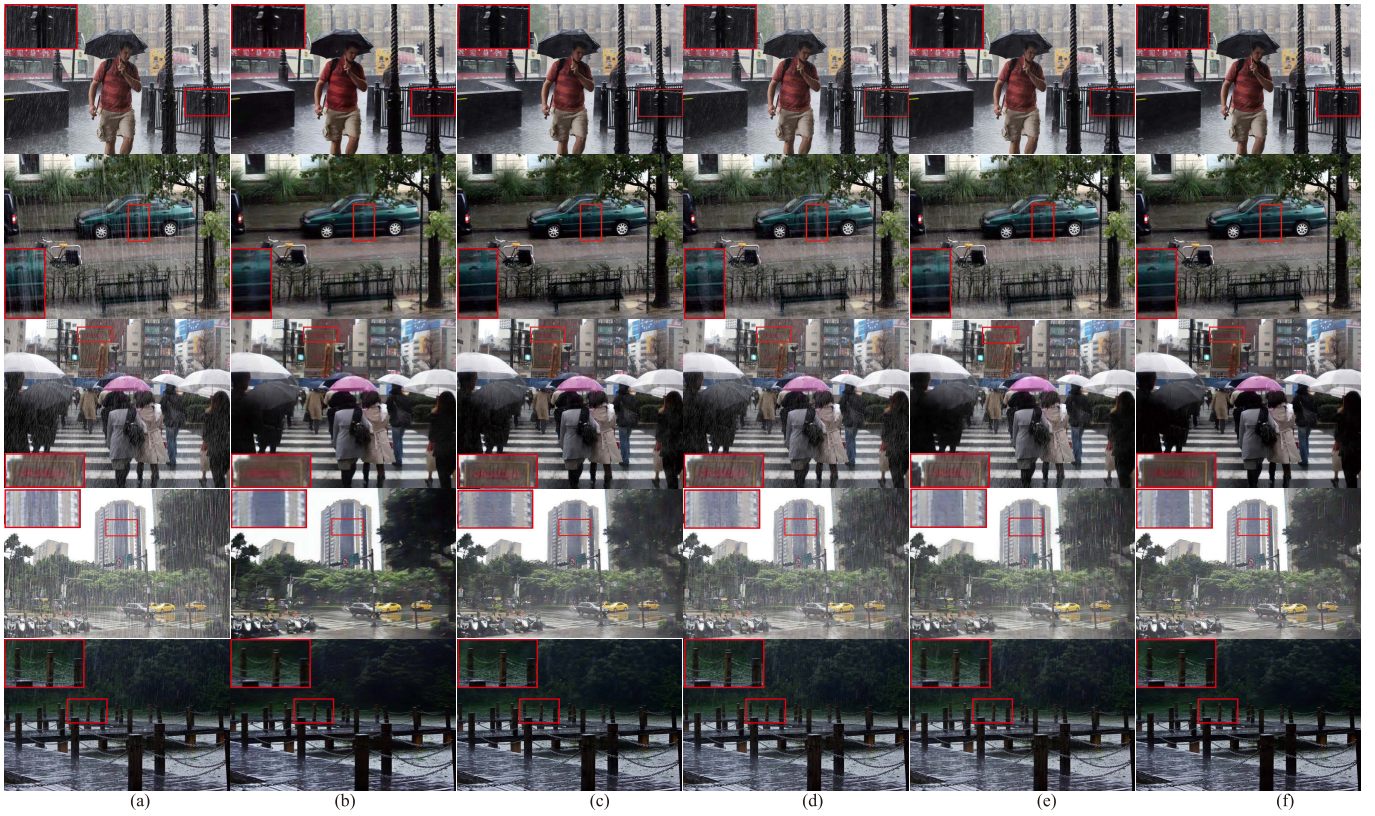


Fig. 10. Visual comparison of real rain removal on Real100rain. (a) Rainy. (b) DID-MDN. (c) DDN. (d) JORDER. (e) SPANet. (f) DRAN.

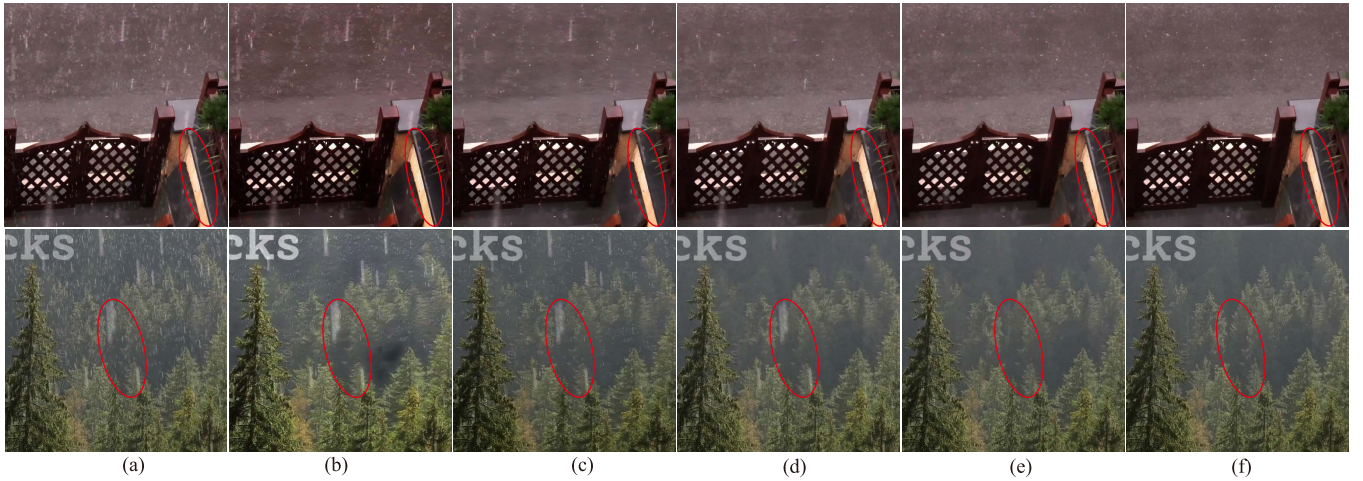


Fig. 11. Visual comparison of real rain RealHQRain [37]. (a) Rainy. (b) DID-MDN. (c) DDN. (d) SPANet. (e) DRAN. (f) Reference.

example, DRAN could better preserve the line pattern iron fence (first row) with less residual rain (second row). In the third row, the text in the zoom region of the proposed method is more clear. The quantitative comparisons on RealHQRain are listed in Table II. We adopt the numerical results reported in [37]. Our DRAN still achieves the best PSNR and SSIM over all the competing methods for the real rain streaks.

We attribute the effectiveness of the proposed method to the curriculum learning framework, in which the angle, rain layer, and clean image are progressively learned in a coarse-to-fine and mutually guidance manner. On one hand, in terms of rain streak modeling, the directionality side information benefits us differing the rain streaks from the line pattern image edges with different angles from the supervised perspective. On the

other hand, in terms of image layer modeling, the RAB origins from the IR, and learns the attention hyperparameters to adaptively enhance the informative image features from the unsupervised perspective. And the joint image and rain streak modeling benefits us with better discriminative representation. That is the main reason why the proposed method is effective for image structure-preserving.

E. Ablation Study

1) *The Effect of Direction*: Fig. 12 gives a visual example of real deraining results with and without the direction-aware network. In Fig. 12(b) for the DRAN without the direction prior, we can observe that the cloth wrinkle (red rectangle) has been unexpectedly suppressed along with the rain streaks.



Fig. 12. Effectiveness analysis of the direction knowledge and RAB. (a) Rainy image. Deraining results (b) without DAN and (c) without RAB. (d) Proposed DRAN.

TABLE III
QUANTITATIVE EVALUATION ON HIGH-LEVEL TASKS

Index	Detection on VOC2007			Segmentation on Cityscape		
	mAP	PSNR	SSIM	mIoU	PSNR	SSIM
GT	0.9547	—	—	0.7839	—	—
Rainy	0.7100	17.06	0.6355	0.1911	28.08	0.7034
Rainy+	0.9448	17.06	0.6355	0.7444	28.08	0.7034
DDN	0.8949	23.98	0.7633	0.6201	36.42	0.9510
JORDER	0.7203	26.71	0.7810	0.5123	32.69	0.8555
SPANet	0.9426	30.24	0.9114	0.6857	37.41	0.9654
DRAN	0.9481	33.67	0.9472	0.7640	40.48	0.9812

TABLE IV
EFFECTIVENESS ANALYSIS OF DIRECTION KNOWLEDGE AND RAB

Direction	RAB	PSNR	SSIM
	✓	34.05	0.9531
✓		33.12	0.9476
✓	✓	34.43	0.9566

TABLE V
INFLUENCE OF THE ESTIMATED ANGLE θ

θ	Estimated	GT	GT $\pm 5^\circ$	GT $\pm 10^\circ$	GT $\pm 20^\circ$	GT $\pm 30^\circ$
PSNR	34.43	34.45	34.40	34.22	33.20	31.09
SSIM	0.9566	0.9566	0.9561	0.9547	0.9421	0.8976

In Fig. 12(d), the DRAN could satisfactorily differentiate the line pattern edges from the rain streaks via the direction-aware discriminative knowledge. Furthermore, Table IV offers a quantitative analysis on rain200Place with and without the direction guidance. The improved 0.38 dB by the direction knowledge validates the effectiveness of DAN.

2) *The Influence of the Angle*: In Table V, we show how the angle estimation influences the deraining performance. We have two key observations. First, if we feed the oracle angle into the network, we can obtain the best deraining results. Moreover, the estimated angle by the proposed method has achieved the results close to that of the GT, which validates the effectiveness of the proposed network. Second, with the lower accuracy of the angle, the deraining results gradually decrease. That is to say, the precise angle guidance can indeed improve the DRAN for better deraining performance. We further validate the estimated angle on real rain examples in Fig. 13. We can intuitively infer that the direction estimation network works well. Moreover, the real rain images have been satisfactorily restored by DRAN in the second row.



Fig. 13. Estimated angle on different real rainy images.

TABLE VI
COMPARISON BETWEEN CONVOLUTION, RESNET, AND RAB

Index	Single Model			Large Model		
	PSNR	SSIM	Parameter	PSNR	SSIM	Parameter
Conv	31.66	0.9350	3.83M	33.12	0.9476	28.1M
Resblock	31.83	0.9353	4.25M	33.16	0.9480	28.1M
RAB	32.80	0.9431	3.78M	34.43	0.9566	28.5M

3) *The Effect of RAB*: Fig. 12 gives a visual example of real deraining results with and without the RAB. In Fig. 12(c), there exist moderate rain streaks (green and yellow rectangles) without RAB. In Fig. 12(d), the DRAN could adaptively enhance the informative feature via the RAB with fewer residuals. Furthermore, Table IV provides a quantitative analysis on rain200Place with and without the RAB. The improved 1.31 dB over Resblock validates the effectiveness of the RAB.

Moreover, we compare the RAB with classical convolution and Resblock in Table VI. We use two baselines: single model and large model. The single model is to validate the effectiveness of each block without any intervene of other factors, which takes the RAN as the baseline deraining model. Each model contains eight conv, Resblocks, and RABs as the basic block. Compared with the single model, the large model takes DRAN as the baseline deraining model. We replace the RAN with the corresponding single model for different blocks. The large model is to further validate the effectiveness of each block with the benefit of the directionality. We can observe that although the parameters of Conv, Resblock, and RAB are approximative, the performance of the RAB-based single and large models is significantly better than that of others.

4) *The Analysis of Learned δ* : We analyze how RAB enhances feature representations with interpretability. Specifically, we illustrate how the learned parameter δ of each RAB dynamically changes in Fig. 14 (red curve). Since the values of each RAB are vectors, we calculate the mean absolute value

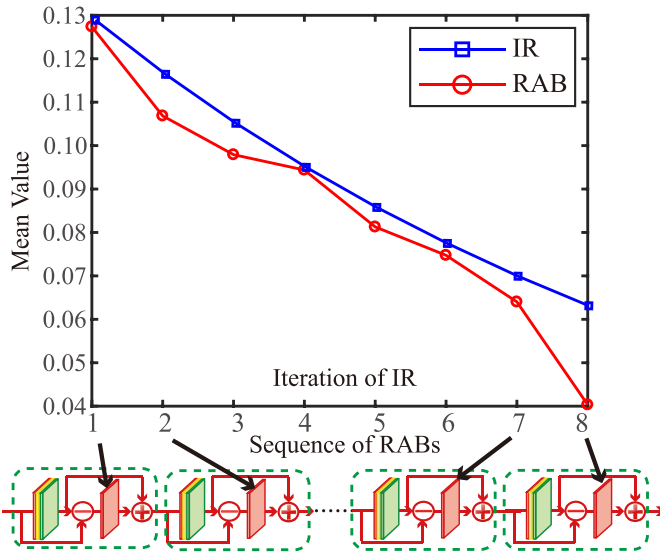


Fig. 14. Correlation between the IR and the RAB. We show the changing curve of the γ over iterations and the learned δ over each RABs.

TABLE VII
EFFECTIVENESS OF THE LEARNED δ

θ	RAB (Learned)	ResBlock	RAB (Fixed)
PSNR	34.43	33.16	33.01
SSIM	0.9566	0.9480	0.9446

of the multiple channels. Interestingly, the learned δ values monotonically decrease with the increasing depths. Moreover, the two curves of the IR and RABs are quite similar to each other. Such a phenomenon is very reasonable. In IR, the decay strategy on the hyperparameter γ is always adopted with the increase in iterations. With the increasing number of RAB, the entangled features between the rain streaks and the image layer are gradually decoupled. Therefore, it is very reasonable to decrease the parameter δ with less retrieval residual. This highlights the fact that the interpretable features in RAB facilitate to satisfactorily remove the rain streaks and well preserve the image details. In turn, we provide a guidance to set the parameter γ for the IR.

To further validate the effectiveness of the learned δ , in Table VII, we take the RABs (adaptive learned), resnet block, and RABs (fixed) as the basic block in RAN, respectively. We set $\delta = 0.5$ for each channel. First, the adaptive learned RAB achieves the best results, which demonstrates the effectiveness of the RAB to attentively learn the discriminative features to better differ the image texture from the residual rain pattern. Moreover, we can observe that once δ has been fixed, the RAB approximately degenerates to the conventional Resblock with similar deraining results, which further validates the close relationship between the RAB and the Resblock. We would like to add this analysis into the main manuscript. Thanks again for the reviewer significantly improving the quality of the work.

5) *The Effect of Curriculum Learning*: The DRAN progressively learns the angle of the rain streak, rain streak layer, and image layer in a coarse-to-fine and curriculum learning way, which is of great benefit to improve feature representation for better deraining result. In Table VIII, we analyze the effect

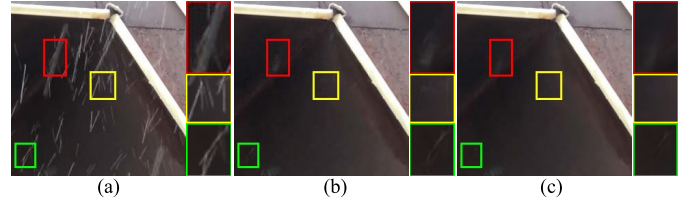


Fig. 15. Robustness to multiple direction real rain streaks. (a) Rainy. (b) DAN. (c) DRAN.

TABLE VIII
EFFECTIVENESS OF SUPERVISION

Direction	Rain Streaks	Image Layer	PSNR	SSIM
		✓	33.62	0.9488
	✓	✓	34.17	0.9548
✓	✓	✓	34.43	0.9566

of each supervision. From the first row to the second row, the additionally learned rain streaks with location and shape information work as the conditional map to facilitate image layer estimation. From the second row to the third row, the direction information further refines the rain streaks so as to indirectly improve image estimation. The progressive guidance via different supervision makes DRAN work well for rain streaks' removal.

6) *The Robustness to Multiple Directions*: Although DAN is designed to use the directionality to improve the performance, our DRAN still works well even if the rain streaks with multiple directions as shown in Fig. 15. Note that the directionality is only auxiliary scalar information to refine rain streaks' estimation. The deraining results still mostly rely on the rain streaks' estimation subnetwork and the residual-aware image layer estimation layer. The DAN obtains the principal angle (5.44°) with the approximate average of all the streaks. In Fig. 15(b), the DAN could still remove most of the rain streaks with the coarse principal angle guidance. The direction prior along with the visual appearance features guarantees the robustness of DAN to multiple direction rain streaks. In Fig. 15(c), the DRAN with the image estimation network could further rectify the result, which further supports the effectiveness of our jointly modeling strategy for both the rain streaks and the image layer.

Note that in Table VIII, these experimental results demonstrate that the image is the baseline, the rain can be a good guidance, and the rain angle is the auxiliary side information to refine rain streaks' estimation. Moreover, in Fig. 13, we have shown that the proposed method does not need the very accurate angle. We can observe that the error ranging between $[-5^\circ, 5^\circ]$ would cause slight influence to the final results, which holds true for most of the real-world rain. Therefore, as for the multiple direction rains, the DRAN can regress its principle direction, and then take the principle direction as guidance with good results. This quantitative result explains why the proposed method is robust to rain streak with multiple directions. The results on the Rain100H dataset with multiple directions also validate the robustness of the proposed method to multiple directional rain streaks.

7) *The Effect of DWT/IDWT*: We use the U-Net as the backbone for rain streaks' appearance modeling. However, the downsampling operators in U-Net may lose useful

TABLE IX
EFFECTIVENESS ANALYSIS OF DWT/IDWT

DWT/IDWT	PSNR	SSIM
	34.04	0.9542
✓	34.43	0.9566

TABLE X
RUNNING TIME COMPARISON (SECONDS). C DENOTES CPU

Size	DSC (C)	GMM (C)	DNN	DID-MDN	JORDER
1024*2048	5066	1443	4.07	1.12	12.16
Size	SPANet	DRDNet	PreNet	JORDER-E	DRAN
1024*2048	10.75	4.78	0.12	8.18	3.58



Fig. 16. Limitation of DRAN for the rainy image with haze. (a) Real rainy image with hazy effect, (b) rain streaks' removal results by DRAN, and (c) MSCNN [85] dehazing + DRAN deraining results.

information of the image, while low-level image restoration may expect to well preserve the pixel-level information. Therefore, we replace the conventional downsampling/upsampling operations with the discrete wavelet transform/inverse discrete wavelet transform (DWT/IDWT), respectively. Since the DWT and IDWT are invertible, we can preserve useful information and enlarge receptive field for the context. In this work, we choose the Haar wavelet. In Table IX, we report the rain streaks' removal results with/without DWT/IDWT. It can be seen that the DWT could indeed improve the deraining performance.

8) *The Running Time*: We report the running time comparison in Table X. The optimization-based GMM and DSC are quite slow on CPU. The learning-based methods are much faster on GPU. Overall, the proposed method DRAN is competitive over other methods, which validates the efficiency of DRAN.

9) *The Limitation*: The joint deraining and dehazing is becoming increasingly popular, since the real rainy weather especially the heavy rain always contains both the rain and the haze. In this work, we mainly focus on the rain streaks without considering the haze in heavy rains in Fig. 16(a). Thus, the DRAN can well handle the rain streaks but fail to remove the haze in Fig. 16(b). However, the proposed DRAN could be easily cascaded with the dehazing network to further handle the real heavy rain. In Fig. 16(c), we show that the restoration result of the cascaded DRAN and MSCNN [85] is a visual pleasure, where both the rain streaks and haze effect of a real heavy rainy scene has been satisfactorily removed.

V. CONCLUSION

In this work, we present a novel DRAN for single-image rain streaks' removal, which enhances the informative feature representation for both rain streaks' modeling and image

layer modeling. On one hand, we incorporate the direction knowledge via DAN to better differentiate the rain streaks from the line pattern image edges; on the other hand, the residual knowledge is further distilled via the RAB to adaptively disentangle the rain streaks from the image details. Moreover, the angle, rain streaks, and image layer in DRAN can mutually complement and reinforce each other in a progressive guidance manner. The solid improvements over the SOTA methods on comprehensive simulated and real datasets demonstrate the effectiveness of DRAN for rain streaks' removal.

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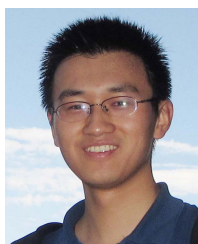
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